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Sufficient dimension reduction and prediction in regression: Asymptotic results

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ABSTRACT

We consider model-based sufficient dimension reduction for generalized linear models and prove the consistency and asymptotic normality of the prediction estimator studied empirically for the normal case by Adragni and Cook (2009) when a sample version of the sufficient dimension reduction is used. Moreover, we provide a formula for the prediction that does need require explicitly computing the reduction.

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1. Introduction

Many statistical problems deal with the study of the conditional distribution of a response $Y \in \mathbb{R}$ given a vector of predictors $\mathbf{X} \in \mathbb{R}^p$, or with the relationship of a set of predictors $\mathbf{X}$ in different populations $Y \in \{1, \ldots, h\}$. When the number of predictors is large, most of the methods used to study these relationships include some type of dimension reduction for $\mathbf{X}$. In particular, if the goal is the estimation of $\mathrm{E}(Y|\mathbf{X})$ or, when Y is categorical, $\Pr(Y = k|\mathbf{X})$ with $k \in \{1, \ldots, h\}$, it is worthwhile to consider predictions based on a function $R(\mathbf{X})$ of dimension less than p, provided that it captures all of the information that $\mathbf{X}$ contains about Y, so that $\mathrm{E}(Y|\mathbf{X}) = \mathrm{E}\{Y|R(\mathbf{X})\}$. In this way no information about $\mathbf{X}$ necessary for the prediction of Y is lost. Among the advantages of dimension reduction is that predictions based on an estimated $R(\mathbf{X})$ may be substantially less variable than those based on $\mathbf{X}$, without introducing bias. Moreover, dimension reduction is particularly useful when it can be performed before modeling the response as a function of predictors, since when the number of predictors is large there are no graphical tools available to do exploratory data analysis to facilitate this modeling process.

Principal component analysis (PCA) is the dominant method of dimension reduction across the applied sciences. It is one of the few tools that is able to get a reduction of $\mathbf{X}$ without first modeling $\mathrm{E}(Y|\mathbf{X})$. Nevertheless, if PCA is used to obtain the reduction $R(\mathbf{X})$, important information on the relation between Y and $\mathbf{X}$ may be lost, affecting future predictions since the principal component reduction does not take into account the response of interest; see [1]. While PCA is most common, another existing method is sufficient dimension reduction, a well establish paradigm of reduction before modeling the regression that does not have this disadvantage of losing information in $\mathbf{X}$ necessary for the prediction of Y.

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In his Fisher lecture, Dennis Cook [11] connected the concept of dimension reduction methodology in regression with the classical statistical concept of sufficiency, and introduced the term Sufficient Dimension Reduction (SDR). He defined a reduction $R : \mathbb{R}^p \to \mathbb{R}^d$ with $d \le p$ as sufficient for the regression of Y on $\mathbf{X}$ if at least one of the following three equivalent statements holds: (i) inverse reduction, $\mathbf{X}|(Y, R(\mathbf{X})) \sim \mathbf{X}|R(\mathbf{X})$; (ii) forward reduction, $Y|\mathbf{X} \sim Y|R(\mathbf{X})$; (iii) joint reduction, $\mathbf{X} \perp\!\!\!\perp Y|R(\mathbf{X})$, where $\perp\!\!\!\perp$ indicates independence and $\sim$ means identically distributed. Each of the three conditions in this definition entails that the reduction $R(\mathbf{X})$ carries all the information that $\mathbf{X}$ has about Y. Most of the existing methodology in SDR is based on the concept of inverse regression, making use of the equivalence (i). The appeal of this approach is due to the fact that it amounts to modeling $\mathbf{X}$ on Y, p one-dimensional regressions, which are much easier to visualize and model than the forward regression of Y on $\mathbf{X}$.

Once an estimator of the SDR, $\hat{R}(\mathbf{X})$, is obtained, we can use graphical tools, if d is small, to model the forward regression Y on $\hat{R}(\mathbf{X})$ or estimate $\mathrm{E}\{Y|R(\mathbf{X})\}$ nonparametrically using $\hat{R}(\mathbf{X})$ instead of $\mathbf{X}$. This is the way sufficient dimension reduction has been successfully used in practice. Nevertheless, since up to now there were no theoretical results to justify this step, i.e., no consistency results or confidence intervals for the prediction, applications of sufficient dimension reduction of this type were justified only on heuristic grounds. The present paper represents the first step towards giving a sound theoretical basis to this exploratory technique.

There are two general paradigms for determining and estimating a sufficient reduction. The first one is the way that sufficient dimension methodology was started and it is based on moments or functions of moments of the conditional distribution of $\mathbf{X}|Y$: SIR [23], SAVE [18], pHd [24], PIR [5], MAVE [15,17,26,34,39], and DR [25]. A defining feature of sufficient reductions obtained via moments is that they are projections of the predictor vector on lower dimensional subspaces and this applies only to continuous predictors. Therefore, sufficient reductions have been by default linear and SDR methodology has mostly been linear SDR.

The second framework developed to determine and estimate SDR works by specifying a parametric model for the inverse regression of $\mathbf{X}$ on Y. In this model-based approach, minimal sufficient reductions can be determined from the model itself; see [1,6,11–14,16,31,32,38]. Under this approach, the SDR, $R(\mathbf{X})$, is not necessarily linear on $\mathbf{X}$ as in the case of moment based methodology; see [6].

Model-based sufficient dimension reduction is relatively new [11]. There are several useful characteristics of this approach, but one of the most remarkable was highlighted by Adragni and Cook [1]: as these authors pointed out, *a model for* $\mathbf{X}|Y$ *can itself be inverted to provide a method for estimating the forward mean function* $\mathrm{E}(Y|\mathbf{X})$ *without specifying a model for the full joint distribution of* (X, Y). This is due to the fact that, using Bayes' Theorem, the regression function $r(x) = \mathrm{E}(Y|\mathbf{X} = x)$ can be written as

$$r(x) = \mathrm{E}\{Yf(x|Y)\}/\mathrm{E}\{f(x|Y)\}, \tag{1}$$

where here, and throughout this article, expressions of the form $f(x|y)$ should be understood as the conditional density, with respect to an appropriate reference measure ν, of $\mathbf{X}|Y = y$ evaluated at x. Implicitly, we assume that there exists a regular conditional probability for $\mathbf{X}|Y = y$, with Radon–Nikodym derivate with respect to ν equal to $f(x|y)$, which we call the conditional density or just density if the meaning is clear from context. Moreover, as stated in [1], if $R(\mathbf{X})$ is a sufficient dimension reduction that has itself a density (or pmf), the regression function can, indeed, be represented as

$$r(x) = \mathrm{E}[Yf\{R(x)|Y\}]/\mathrm{E}[f\{R(x)|Y\}], \tag{2}$$

where $f\{R(x)|Y\}$ denotes now the density of $R(\mathbf{X})$ given Y, evaluated at $R(x)$. Eq. (2) provides a new method to estimate $r(x)$ as

$$\widehat{r}_n^{\mathrm{res}}(x) = \sum_{i=1}^n w_i^{\mathrm{res}}(x)Y_i \quad \text{with} \quad w_i^{\mathrm{res}}(x) = \hat{f}\{\hat{R}(x)|Y_i\}\Big/\sum_{j=1}^n \hat{f}\{\hat{R}(x)|Y_j\}, \tag{3}$$

where $\hat{f}(\cdot|y)$ denotes an estimator of the conditional density of $R(\mathbf{X})|Y = y$ and $\hat{R}(x)$ is the estimated reduction. Adragni and Cook [1] proved that the estimator given by (3) when $\mathbf{X}|Y = y \sim \mathcal{N}(\mu_y, \Delta)$ has an excellent behavior in simulations and data analysis. Nonetheless the question that remains open is the study of the theoretical properties of such an estimator, namely, establishing its consistency and asymptotic behavior.

The goal of this paper is to answer this question when $\mathbf{X}|Y$ follows a generalized linear model. For this inverse regression model, Bura et al. [6] characterized the SDR, $R(\mathbf{X})$, for the regression of Y on $\mathbf{X}$. Let us note that the case considered in [1] is a particular example of such a model. Interestingly, there are some remarkable results that underlie the proof of the asymptotic behavior of $\widehat{r}_n^{\mathrm{res}}(x)$ that are not only of theoretical interest, but are also useful for the practical implementation of (3). To briefly discuss such results, we have to come back to the representation (1) that allows the estimation of the regression $r(x)$ (even before talking about reduction) as

$$\widehat{r}_n(x) = \sum_{i=1}^n w_i(x)Y_i \quad \text{with } w_i(x) = \hat{f}(x|Y_i)\Big/\sum_{j=1}^n \hat{f}(x|Y_j). \tag{4}$$

Now, under the assumed inverse-model framework, the shape of f is known, up to the value of its parameter ξ_0. Namely, if we assume that $\mathbf{X}|Y = y \sim f(\cdot|y, \xi_0)$, we can choose

$$w_i(x) = f(x|Y_i, \widehat{\xi}_n) \Big/ \sum_{j=1}^{n} f(x|Y_j, \widehat{\xi}_n), \tag{5}$$

where $\widehat{\xi}_n$ is any consistent estimator of ξ_0.

The first results of this paper deal with the asymptotic behavior of $\widehat{r}_n(x)$; see Section 2.3. We will establish the required conditions to deduce the consistency of $\widehat{r}_n(x)$ for $r(x)$ when $\widehat{\xi}_n$ is a consistent estimator of ξ_0. Moreover, asymptotic normality of $\sqrt{n}\,\{\widehat{r}_n(x) - r(x)\}$ will follow from the asymptotic linear expansion of $\sqrt{n}\,(\widehat{\xi}_n - \xi_0)$. Of course (4) does not take into account that we could have a reduction $R(\mathbf{X})$, and therefore we could have a more efficient estimator of $r(x)$. As we already mentioned, Bura et al. [6] showed a SDR, $R(\mathbf{X})$, for the case of $\mathbf{X}|Y$ following a generalized linear model. The existence of a non trivial reduction ($d < p$) implies that the parameter ξ_0 of the generalized linear model belongs to a lower dimensional space.

Estimating ξ_0 with a consistent estimator $\widehat{\xi}_n^{\text{res}}$ belonging to the assumed low-dimensional space allows us to get another estimator of $r(x)$ if we replace the weights given in (5) by the weights

$$w_i(x) = f(x|Y_i, \widehat{\xi}_n^{\text{res}}) \Big/ \sum_{j=1}^{n} f(x|Y_j, \widehat{\xi}_n^{\text{res}}); \tag{6}$$

see Section 3. The interesting property that we will prove later is that the weights given in (6) are equal to the weights defined in (3).

As a consequence, the consistency and asymptotic normality of $\widehat{r}_n^{\text{res}}(x)$ will be deduced from that of $\widehat{r}_n(x)$. This theoretical result gives a way to implement $\widehat{r}_n^{\text{res}}(x)$ avoiding the estimation of a reduction and its conditional distribution, by computing the weights according to the formula (6). In Section 4 we show that, as expected, when maximum likelihood estimators of ξ_0 are imputed, the weights defined in (6) give a more efficient estimator than the one given in (3) and as a result $\widehat{r}_n^{\text{res}}(x)$ will be a more efficient estimator of the regression function $r(x)$ than $\widehat{r}_n(x)$. In Section 5, we report the results of a simulation study. In Section 6, using an example considered in [6], we show how to compute in practice the approximate confidence intervals for the mean as well as for the future prediction.

2. Asymptotic behavior of a regression estimator under a multivariate generalized linear model

2.1. Exponential families

In this paper we consider the estimation of the regression function $r(x) = \mathrm{E}(Y|\mathbf{X} = x)$ using $(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)$ independent, identically distributed (iid) as $(\mathbf{X}, Y)$. As stated in the Introduction, one of the main goals is to analyze the asymptotic behavior of the estimator $\widehat{r}_n^{\text{res}}(x)$ defined in (3) when a generalized linear model is imposed for the inverse regression of $\mathbf{X}|Y = y$, as in [6].

To do so, we need first to recall the definition of an exponential family. Let $\mathbf{X} = (X_1, \ldots, X_p)^\top$ be a p-dimensional random vector and assume that its distribution belongs to a k-parameter canonical exponential family with density function (with respect to an appropriate reference measure)

$$f_\eta(x) = \exp\{\eta^\top \mathbf{T}(x) - \psi(\eta)\} h(x), \tag{7}$$

where $\mathbf{T}(x) = (t_1(x), \ldots, t_k(x))^\top$ is a vector of known real-valued functions, $h(x) \geq 0$ is a non-negative known function and $\eta \in \mathbb{R}^k$ is the vector of natural parameters, taking values in

$$H = \left\{ \eta \in \mathbb{R}^k : \int \exp\{\eta^\top \mathbf{T}(z)\} h(z) dz < \infty \right\},$$

where the integral is replaced by a sum when $\mathbf{X}$ is discrete. The set H of natural parameters is assumed to be open and convex in $\mathbb{R}^k$. Moreover, we assume that $\psi(\eta)$ is strictly convex and infinitely differentiable in H. In particular, this implies $\mathrm{E}_\eta\{t_s(\mathbf{X})\} = \partial\psi(\eta)/\partial\eta_s$, $\mathrm{cov}_\eta\{t_s(\mathbf{X}), t_j(\mathbf{X})\} = \partial^2\psi(\eta)/\partial\eta_s\partial\eta_j$. Since ψ is strictly convex, $\mathrm{var}_\eta\{\mathbf{T}(\mathbf{X})\}$ is non-singular for every $\eta \in H$. Furthermore, we assume that $\mathbf{T}$ is a minimal complete and sufficient statistics for the family. A thorough treatment of the exponential family is given in [3].

Exponential families include, for example, the multivariate normal model, the multivariate Bernoulli distributions, as considered in [19], or the Ising multivariate Bernoulli distributions, as discussed in [9]. We can also consider a mixed family, as in [8,35]. An example of such a model that will be used in Section 6 is obtained when a Bernoulli experiment is followed by a normally distributed one. In general, the Bernoulli variable is used to represent two levels of a categorical variable (such as gender) while the normal distribution is used to model a vector of measurements whose distribution varies between the two categories. Namely, consider $\mathbf{X} = (W, \mathbf{Z}) \in \mathbb{R}^p$, where W follows a Bernoulli distribution with parameter α and $\mathbf{Z}$ is a $(p-1) \times 1$ random vector whose conditional distribution given $W = w$ is normal with mean $\mu + bw$ and covariance matrix Σ, which does not depend on w; i.e., $\mathbf{Z}|W = w \sim \mathcal{N}(\mu + bw, \Sigma)$. Then, defining $\eta = (\eta_1, \eta_2^\top, \eta_3^\top, \eta_4^\top)^\top$, with

$$\eta_1 = \ln\left(\frac{\alpha}{1-\alpha}\right) - \frac{1}{2} b^\top \Sigma^{-1} b - b^\top \Sigma^{-1} \mu\,, \quad \eta_2 = \Sigma^{-1}\mu, \quad \eta_3 \Sigma^{-1} b\,, \quad \eta_4 = \mathrm{vech}(\Sigma^{-1}), \tag{8}$$

and $\mathbf{T}(x) = (T_1(w,z), \mathbf{T}_2^\top(w,z), \mathbf{T}_3^\top(w,z), \mathbf{T}_4^\top(w,z))^\top$, where

$$T_1(w,z) = w, \quad \mathbf{T}_2(w,z) = z\ , \quad \mathbf{T}_3(w,z) = wz\ , \quad \mathbf{T}_4(w,z) = -\frac{1}{2} D_{p-1}^\top D_{p-1}\mathrm{vech}(zz^\top),$$

we have that for, some strictly convex function ψ of η,

$$f_{W,\mathbf{Z}}(w,z) = f_\eta(w,z) = \exp\{\eta^\top \mathbf{T}(x) - \psi(\eta)\},$$

where $\mathrm{vec}(\mathbf{A}) \in \mathbb{R}^{q^2}$ indicates the vectorization by column of a matrix $\mathbf{A} \in \mathbb{R}^{q\times q}$, $\mathrm{vech}(\mathbf{A}) \in \mathbb{R}^{q(q+1)/2}$ is the half vectorization of a symmetric matrix $\mathbf{A}$ considering only the elements of the diagonal and upper diagonal of the symmetric matrix $\mathbf{A}$ and D_q is the duplication matrix such that for $\mathbf{A}$ symmetric, $\mathrm{vec}(\mathbf{A}) = D_q\mathrm{vech}(\mathbf{A})$.

2.2. Multivariate generalized linear models when $\mathbf{X}$ and Y are both stochastic

In our setting, we also have a vector Y that is observed and, throughout, we assume that $(\mathbf{X}, Y)$ follows an inverse generalized linear model (inverse GLM); see [27,29,30]. Such models postulate that the conditional density of $\mathbf{X}|Y = y$ belongs to some fixed exponential family and hypothesizes that the k-vector of natural parameters, that we will call now η_y to emphasize the dependence on y, depends linearly on ν_y, an r-valued vector of some known functions of y with $\mathrm{E}(\nu_Y) = 0$. Frequently, a subset of the natural parameters depends on y while its complement does not. The normal linear model with constant variance is such an example. To accommodate this structure, we partition the vector η_y indexing model (7) in η_{y1} and η_{y2}, with k_1 and k_2 components, respectively, and we assume that H, the natural parameter space of the exponential family, equals $\mathbb{R}^{k_1} \times H_2$, where H_2 is an open convex subset of $\mathbb{R}^{k_2}$. Thus, $\eta_y = \eta_\xi(y)$, with

$$\eta_\xi(y) = \begin{pmatrix} \eta_{y1} \\ \eta_{y2} \end{pmatrix} = \begin{pmatrix} \overline{\eta}_1 + \beta\ \nu_y \\ \overline{\eta}_2 \end{pmatrix}, \tag{9}$$

where $\xi = (\overline{\eta}_1^\top, \mathrm{vec}^\top(\beta), \overline{\eta}_2^\top)^\top \in \Xi = \mathbb{R}^{k_1} \times \mathbb{R}^{k_1 r} \times H_2$, denotes a generic vector and ξ_0 the true parameter. Note that $\Xi \subseteq \mathbb{R}^K$, with $K = k_1(1+r) + k_2$. In what follows, to keep the notation as simple as possible, we concatenate columns vectors omitting transposing them; i.e., we use (a, b) for $(a^\top, b^\top)^\top$. Moreover, we use $\xi = \left(\overline{\eta}_1, \beta, \overline{\eta}_2\right)$, understanding here that β stands for $\mathrm{vec}(\beta)$. Let $f(\cdot|y, \xi)$ denotes the conditional density of $\mathbf{X}|Y = y$ when an exponential family model is assumed for the conditional distribution of $\mathbf{X}|Y = y$, as in (7), and the natural parameter follows the linear model postulated in (9).

2.3. Regression estimation and asymptotic results

The representation of the regression function provided in Eq. (1), indicates that

$$r(x) = \mathrm{E}\{Yf(x|Y, \xi_0)\}/\mathrm{E}\{f(x|Y, \xi_0)\}.$$

Notice that both expectations are taken with respect to the marginal distribution of Y, and this will be the case in any instance where x is fixed and only Y is involved in the argument of the expectation. Combining (7) with (9), the regression function can, indeed, be written as

$$r(x)xx = \mathrm{E}\{g_{1,x,\xi_0}(Y)\} \Big/ \mathrm{E}\{g_{0,x,\xi_0}(Y)\}, \tag{10}$$

defining

$$g_{j,x,\xi}(y) = y^j \exp\{(\overline{\eta}_1 + \beta\nu_y)^\top \mathbf{T}_1(x) + \overline{\eta}_2^\top \mathbf{T}_2(x) - \psi(\overline{\eta}_1 + \beta\nu_y, \overline{\eta}_2)\}, \tag{11}$$

where $\mathbf{T}_1(x)$ contains the first k_1 components of $\mathbf{T}(x)$, while $\mathbf{T}_2(x)$ is the vector with the remaining k_2 coordinates. This expression of the regression function suggests that it can be estimated through a plug-in procedure, replacing in (10) the true parameter ξ_0 with an estimator $\widehat{\xi}_n$ and the mean operator by an empirical one, giving rise to the following estimator

$$\widehat{r}_n(x) = \mathrm{P}_n\{g_{1,x,\widehat{\xi}_n}(Y)\} \Big/ \mathrm{P}_n\{g_{0,x,\widehat{\xi}_n}(Y)\}, \tag{12}$$

where P_n denotes the empirical mean operator; namely $\mathrm{P}_n\{g(Y)\} = \{g(Y_1) + \cdots + g(Y_n)\}/n$. Interestingly, the estimator $\widehat{r}_n(x)$ is indeed a weighted average $w_1(x)Y_1 + \cdots + w_n(x)Y_n$, where the weights are defined as in (5).

We will show in Theorem 1, that, under integrability conditions stated in (13), if $\widehat{\xi}_n$ is a consistent estimator of ξ_0, the numerator and denominator of (12) converge to $\mathrm{E}\{g_{j,x,\xi_0}(Y)\}$ for $j = 1$ and for $j = 0$, respectively, and therefore, $\widehat{r}_n(x)$ converges to $r(x)$. This result will be deduced by calling on arguments involving uniform laws of large numbers. More precisely, we will argue that $\{g_{j,x,\xi} : \xi \in \mathcal{K}_0\}$, for $j \in \{0, 1\}$ and $\mathcal{K}_0$ being a compact set containing the true parameter ξ_0 in its interior, is a Glivenko–Cantelli (GC) class, provided that each class has an envelope with finite mean.

Theorem 1. *Let $(\mathbf{X}, Y)$ follow an inverse GLM and let $(\mathbf{X}_1, Y_1), (\mathbf{X}_2, Y_2), \ldots$ be an iid sequence, distributed as $(\mathbf{X}, Y)$. Let $\widehat{\xi}_n$ denote a sequence of estimators computed from $(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)$ that converges to ξ_0 in probability. Suppose that there exists a compact set $\mathcal{K}_0$ with $\xi_0 \in \mathcal{K}_0^o$, the interior of $\mathcal{K}_0$, such that, for $j \in \{0, 1\}$,*

$$\mathrm{E}\left\{\sup_{\xi\in\mathcal{K}_0} \left\|g_{j,x,\xi}(Y)\right\|\right\} < \infty, \tag{13}$$

for $g_{j,x,\xi}$ with $j \in \{0, 1\}$, defined in Eq. (11). Then $\widehat{r}_n(x)$ converges to $r(x)$ in probability, whenever $f(x) > 0$, where f denotes the density function of X.

The next result provides a linear expansion for the estimator $\widehat{r}_n(x)$ when such an expansion exists for $\widehat{\xi}_n$. As a consequence, we can deduce that the estimator $\widehat{r}_n(x)$ is $\sqrt{n}$ asymptotically normal.

Theorem 2. *Let $(\mathbf{X}, Y)$ follow an inverse GLM and let $(\mathbf{X}_1, Y_1), (\mathbf{X}_2, Y_2), \ldots$ be an iid sequence, distributed as $(\mathbf{X}, Y)$. Assume that $f(x) > 0$. Let $\widehat{\xi}_n$ denote a sequence of estimators of ξ_0 computed from $(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)$, satisfying a linear expansion*

$$\sqrt{n}\,(\widehat{\xi}_n - \xi_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \ell(\mathbf{X}_i, Y_i) + o_P(1), \tag{14}$$

where $\mathrm{E}\{\ell(\mathbf{X}, Y)\} = 0$ and $\mathrm{E}\{\|\ell(\mathbf{X}, Y)\|^2\} < \infty$. Assume also that there exists a compact set $\mathcal{K}_0$ with $\xi_0 \in \mathcal{K}_0^o$ such that, for $j \in \{0, 1\}$,

$$\mathrm{E}\left\{\sup_{\xi\in\mathcal{K}_0} \left\|\frac{\partial}{\partial\xi^\top} g_{j,x,\xi}(Y)\right\|\right\} < \infty, \tag{15}$$

where $\partial g_{j,x,\xi}/\partial\xi^\top$ denotes the row vector of derivatives with respect to each coordinate of the vector ξ. Then the following asymptotic expansion for $\widehat{r}_n(x)$ holds

$$\sqrt{n}\,\{\widehat{r}_n(x) - r(x)\} = \frac{1}{C}\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left[\{Y_i - r(x)\}\, g_{0,x,\xi_0}(Y_i) + D\ell(\mathbf{X}_i, Y_i)\right] + o_P(1)$$

with

$$C = \mathrm{E}\{g_{0,x,\xi_0}(Y)\}, \quad D = \mathrm{E}\left[\{Y - r(x)\}\, \frac{\partial}{\partial\xi^\top} g_{0,x,\xi_0}(Y)\right]. \tag{16}$$

Moreover, $\sqrt{n}\,\{\widehat{r}_n(x) - r(x)\}$ is asymptotically normal with mean zero and asymptotic variance

$$avar[\sqrt{n}\,\{\widehat{r}_n(x) - r(x)\}] = C^{-2}\mathrm{E}\left[[\{Y - r(x)\}g_{0,x,\xi_0}(Y) + D\ell(\mathbf{X}, Y)]^2\right].$$

Remark 1. Bura et al. [7] showed that the sequence of maximum likelihood estimators (MLE) of ξ_0 converges in probability to ξ_0 and satisfies (14). In Section 4 we review the conditions required for such results. Therefore, the consistency and asymptotic normality of $\widehat{r}(x)$, defined in (12) follows from choosing as $\widehat{\xi}_n$ the sequence of the MLE of ξ_0.

Remark 2. Theorems 1 and 2 are true in greater generality. If $r(x)$ can be represented as in (10), the only requirement for the consistency of $\widehat{r}_n(x)$, beside that of the imputed estimator $\widehat{\xi}_n$, is that the corresponding class of functions $g_{j,x,\xi}$ are GC and verify property (13). For the asymptotic expansion, also condition (15) needs to be assumed, as well as a linear expansion for $\widehat{\xi}_n$.

3. Reduced rank generalized linear models

Recall that we are modeling the inverse regression of $\mathbf{X}$ on Y and, therefore, in this context $\mathbf{X}$ plays the role of a multivariate response. To motivate the definition of a reduced rank generalized linear model, consider first a multivariate linear regression model where the coordinates of the response vector $\mathbf{X}$ are correlated. This information can be incorporated into the model if we admit the possibility that the regression matrix β does not have full rank. This type of constraint in the regression parameter, which leads to more efficient estimators, was introduced in [2] and [22] in the classical linear regression setting and they were extended not long ago to the exponential family regression model (7) with (9) in [7,36,37]. In the setting of reduced rank generalized linear model (RRGLM), the β-component of the true ξ_0 presented in (9) is in $\mathbb{R}_d^{k_1\times r}$, the set of matrices in $\mathbb{R}^{k_1\times r}$ of rank d for known $d \le \min(k_1, r)$. When the β-component of ξ satisfies the rank restriction, we will say that $\xi \in \Xi^{\mathrm{res}}$.

Bura et al. [6] characterized the minimal sufficient reduction for Y on $\mathbf{X}$ when $(\mathbf{X}, Y)$ follows an inverse reduced rank generalized linear model, assuming that $\beta \in \mathbb{R}_d^{k_1\times r}$. In such a case, $\beta = \mathbf{A}B$ with $\mathbf{A} \in \mathbb{R}^{k_1\times d}$ semi-orthogonal and $B \in \mathbb{R}^{d\times r}$. The authors proved that $R(\mathbf{X}) = \mathbf{A}^\top\mathbf{T}_1(\mathbf{X})$ is a minimal sufficient dimension reduction; see Theorem 1 in [6]. Moreover any $R(\mathbf{X}) = M\mathbf{A}^\top\mathbf{T}_1(\mathbf{X})$, with $M \in \mathbb{R}^{d\times d}$ an invertible matrix, is a minimal sufficient dimension reduction for the regression of

Y on $\mathbf{X}$. Notice that we could use $R(\mathbf{X}, \xi)$ to denote a reduction under the inverse reduced rank generalized linear model with parameter ξ. However, we will omit the ξ-dependence in the remainder of this presentation when we refer to R. For instance, when $\mathbf{X}|Y = y \sim f(\cdot|y, \xi)$ with $\xi = (\bar{\eta}_1, \beta, \bar{\eta}_2)$ with $\beta = \mathbf{A}B$, we use $f\{R(x)|y, \xi\}$ to denotes the conditional density of the reduction $R(\mathbf{X}, \xi)$ given $Y = y$, evaluated at $R(x, \xi)$.

Once the reduction is characterized, we can take advantage of it to estimate the regression function following the approach suggested in [1] and presented in (3). To do so, let $\widehat{\xi}_n^{\text{res}}$ denote an estimator of ξ_0 under the rank restriction model: $\widehat{\xi}_n^{\text{res}} \in \Xi^{\text{res}}$; therefore, its β-component satisfies the rank restriction and, thus, it can be written as $\widehat{\beta}_n^{\text{res}} = \hat{\mathbf{A}}\hat{B}$. From [6] we know that $\hat{\mathbf{A}}^\top \mathbf{T}_1(\mathbf{X})$ is a reduction for the regression of Y on $\mathbf{X}$ when $\mathbf{X}|Y = y \sim f(\cdot|y, \widehat{\xi}_n^{\text{res}})$ and thus $\hat{R}(\mathbf{X}) = \hat{\mathbf{A}}^\top \mathbf{T}_1(\mathbf{X})$ can be considered as an estimated reduction. Therefore, according to (3), the regression function under the inverse RRGLM can be estimated by

$$\widehat{r}_n^{\text{res}}(x) = \sum_{i=1}^{n} w_i^{\text{res}}(x) Y_i, \tag{17}$$

where

$$w_i^{\text{res}}(x) = f\{\hat{\mathbf{A}}^\top \mathbf{T}_1(x)|Y_i, \widehat{\xi}_n^{\text{res}}\} \Big/ \sum_{j=1}^{n} f\{\hat{\mathbf{A}}^\top \mathbf{T}_1(x)|Y_j, \widehat{\xi}_n^{\text{res}}\}. \tag{18}$$

It is known that the reduction is not unique since $M\mathbf{A}^\top \mathbf{X}$ with $M \in \mathbb{R}^{d\times d}$ invertible is also a minimal sufficient reduction. Therefore, $\hat{R}$ is not unique either. However, $\widehat{r}_n^{\text{res}}(x)$ remains the same if we change $\hat{\mathbf{A}}$ by $\hat{\mathbf{A}}M$ with $M \in \mathbb{R}^{d\times d}$ invertible as it will be proven in Lemma 1.

The following result constitutes one of the main contributions of the present work. It states the consistency and the asymptotic normality of the regression estimator suggested in [1], based on the estimated reduction, once the weights are chosen as in (18), for any consistent $\widehat{\xi}_n^{\text{res}} \in \Xi^{\text{res}}$ satisfying a linear expansion when $(\mathbf{X}, Y)$ follows an inverse RRGLM.

Theorem 3. *Let $(\mathbf{X}, Y)$ follow an inverse RRGLM and let $(\mathbf{X}_1, Y_1), (\mathbf{X}_2, Y_2), \ldots$ be an iid sequence, distributed as $(\mathbf{X}, Y)$. Assume that $f(x) > 0$. Let $\widehat{\xi}_n^{res}$ denote a sequence of estimators computed from $(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)$ that converges to ξ_0 in probability, such that $\widehat{\xi}_n^{res}$ belongs to Ξ^{res}. Consider the regression estimator $\widehat{r}_n^{\text{res}}(x)$, defined in (17), using the weight w_i^{res} presented in (18). Assume that condition (13) is satisfied. Then, $\widehat{r}_n^{\text{res}}(x)$ converges in probability to $r(x)$. Suppose now that $\widehat{\xi}_n^{res}$ satisfies the following linear expansion*

$$\sqrt{n}\,(\widehat{\xi}_n^{res} - \xi_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \ell^{\text{res}}(\mathbf{X}_i, Y_i) + o_P(1).$$

where $\mathrm{E}\{\ell^{\text{res}}(\mathbf{X}, Y)\} = 0$ and $\mathrm{E}\{\|\ell^{\text{res}}(\mathbf{X}, Y)\|^2\} < \infty$. Assume that condition (15) is satisfied. Then,

$$\sqrt{n}\left\{\widehat{r}_n^{\text{res}}(x) - r(x)\right\} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left[\{Y_i - r(x)\}\, g_{0,x,\xi_0}(Y_i) + D\ell^{\text{res}}(\mathbf{X}_i, Y_i)\right] + o_P(1),$$

with C and D defined in (16), and $\sqrt{n}\,\{\widehat{r}_n^{\text{res}}(x) - r(x)\}$ is asymptotically normal with mean zero and asymptotic variance

$$avar\left[\sqrt{n}\left\{\widehat{r}_n^{\text{res}}(x) - r(x)\right\}\right] = C^{-2}\mathrm{E}([\{Y - r(x)\}g_{0,x,\xi_0}(Y) + D\ell^{\text{res}}(\mathbf{X}, Y)]^2).$$

As shown in [7], the sequence of maximum likelihood estimators of ξ_0 under the inverse RRGLM satisfies the conditions required in Theorem 3. Therefore the consistency and asymptotic normality of $\widehat{r}_n^{\text{res}}(x)$, defined in (17), follows from choosing such a sequence $(\widehat{\xi}_n^{\text{res}})_1, (\widehat{\xi}_n^{\text{res}})_2, \ldots$ Moreover, in Theorem 4 of Section 4, the asymptotic variances of both $\widehat{r}_n(x)$ and $\widehat{r}_n^{\text{res}}(x)$ are established when MLE are used, and also compared. As expected, the asymptotic variance of $\widehat{r}_n^{\text{res}}(x)$ is smaller than that of $\widehat{r}_n(x)$, exhibiting the announced reduction in variability when the reduction is used to predict Y.

It is worth mentioning that, as it is known, even if the reduction $R(\mathbf{X})$ is not unique, the proof of Theorem 3 does not use any explicit representation nor the consistency of $\hat{R}(\mathbf{X})$. In fact, it only relies on the consistency of the estimator of the identifiable parameter $\xi_0 \in \Xi^{\text{res}}$ in the reduced rank multivariate generalized linear model. This is due to the following result.

Lemma 1. *Assume that $R(\mathbf{X})$ is a sufficient dimension reduction for the regression of Y on $\mathbf{X}$. Then, for any y_1 and y_2, we have that $f(x|y_1)/f(x|y_2) = f\{R(x)|y_1\}/f\{R(x)|y_2\}$.*

Lemma 1 implies that the plug-in estimator $\widehat{r}_n^{\text{res}}(x)$ estimates the regression function $r(x)$ as indicated in (12), but inputting now $\widehat{\xi}_n^{\text{res}} \in \Xi^{\text{res}}$ for estimating ξ_0. Namely, $\widehat{r}_n^{\text{res}}(x)$ is equal to the estimator defined on (17) with the weights $w_i^{\text{res}}(x)$ as defined in (5), substituting $\hat{\xi}_n$ by $\widehat{\xi}_n^{\text{res}}$. Therefore, in practice, to use the estimator $\widehat{r}_n^{\text{res}}(x)$ we do not need to choose $\hat{R}(\mathbf{X})$ nor to get the density for $R(\mathbf{X})|Y$, i.e.,

$$\widehat{r}_n^{\text{res}}(x) = \sum_{i=1}^{n} Y_i f(x|Y_i, \widehat{\xi}_n^{\text{res}}) \Big/ \sum_{i=1}^{n} f(x|Y_i, \widehat{\xi}_n^{\text{res}}).$$

4. Asymptotic behavior of $\widehat{r}_n(x)$ and of $\widehat{r}_n^{\text{res}}(x)$ based on maximum likelihood estimates of ξ_0

The consistency of both $\widehat{r}_n(x)$ and of $\widehat{r}_n^{\text{res}}(x)$ established in Theorems 1 and 2, together with Theorem 3, assume that ξ_0 is consistently estimated by $\widehat{\xi}_n \in \Xi$, and with $\widehat{\xi}_n^{\text{res}} \in \Xi^{\text{res}}$ in the restricted rank model. Furthermore, the asymptotic normality of each procedure assumed that the inputted estimator of ξ_0 satisfies a linear expansion. None of these results refers to any particular estimator, as long as the mentioned properties are guaranteed. Such is the case when the MLE is considered, as stated in [7] under some integrability conditions presented below.

At this point we understand that it is worth investing in improving the notation in order to simplify the presentation of such conditions. Recall that $K = k_1(r+1) + k_2$, and for $z \in \mathbb{R}^r$, let $F(z)$ be the $k \times K$ matrix defined by

$$F(z) = \begin{pmatrix} (1, z^\top) \otimes \mathbf{I}_{k_1} & 0 \\ 0 & \mathbf{I}_{k_2} \end{pmatrix},$$

and so, model (9) can be rewritten as $\eta_y = F(\nu_y)\xi$.

Conditions (19)–(21) stated below together with conditions (13) and (15) are required in [7] to prove the consistency and asymptotic distribution of the MLE estimator of ξ_0 under the GLM presented in Section 2.1 and the corresponding RRGLM presented in Section 3, except for the fact that $\mathbf{X}$ and Y are exchanged. Thus, combining the results presented in [7] and those included in Sections 2.3 and 3, we will establish the consistency and asymptotic normality of $\widehat{r}_n(x)$ and $\widehat{r}_n^{\text{res}}(x)$ if the unrestricted and the restricted MLE of ξ_0 are considered, respectively. We will first state the regularity conditions required to derive consistency and asymptotic distribution of the MLE. First,

$$\Pr\{F(\nu_Y)\xi = \overline{\eta}\} < 1 \text{ for any } \xi \text{ and any } \overline{\eta}. \tag{19}$$

Next, suppose that there exists a compact set $\mathcal{K}_0$ with $\xi_0 \in \mathcal{K}_0^o$, the interior of $\mathcal{K}_0$, such that

$$\mathrm{E}\{\|\mathbf{T}(\mathbf{X})^\top F(\nu_Y)\|\} < \infty, \quad \mathrm{E}\{\|\mathbf{T}(\mathbf{X})^\top F(\nu_Y)\|^2\} < \infty, \tag{20}$$

$$\mathrm{E}\left[\sup_{\xi\in\mathcal{K}_0} |\psi\{F(\nu_Y)\xi\}|\right] < \infty, \quad \mathrm{E}\left[\sup_{\xi\in\mathcal{K}_0} \|\nabla\psi\{F(\nu_Y)^\top\xi\}F(\nu_Y)\|^2\right] < \infty, \quad \mathrm{E}\left[\sup_{\xi\in\mathcal{K}_0} \|F(\nu_Y)^\top\nabla^2\psi\{F(\nu_Y)\xi\}F(\nu_Y)\|\right] < \infty, \tag{21}$$

where $\nabla\psi$ and $\nabla^2\psi$ stand for the $1 \times k$ vector of partial derivatives of ψ and the $k \times k$ symmetric matrix of second order derivatives of ψ, respectively.

We can now show that, as announced, estimating the regression function based on a reduction using MLE of ξ_0 results in a less variable procedure; see Remark 3.

Theorem 4. *Assume that $(\mathbf{X}, Y)$ follows an inverse GLM with true parameter $\xi_0 \in \Xi$ and let $(\mathbf{X}_1, Y_1), (\mathbf{X}_2, Y_2), \ldots$ be an iid sequence, distributed as $(\mathbf{X}, Y)$. Assume that $f(x) > 0$. Choose $\widehat{\xi}_n$ being the maximum likelihood estimate of ξ_0, computed from $(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)$. Then, under conditions (13), (15), (19), (20) and (21), the following asymptotic distribution holds*

$$avar\left[\sqrt{n}\,\{\widehat{r}_n(x) - r(x)\}\right] = C^{-2}\mathrm{E}\left[[\{Y - r(x)\}g_{0,x,\xi_0}(Y)]^2\right] + C^{-2}DV_{\xi_0}^{-1}D^\top, \tag{22}$$

where C and D defined in (16) and

$$V_{\xi_0} = \mathrm{E}[F(\nu_Y)^\top\nabla^2\psi\{F(\nu_Y)\xi_0\}F(\nu_Y)]. \tag{23}$$

Furthermore, if ξ_0 belongs to Ξ^{res} and we take $\widehat{\xi}_n^{res} \in \Xi^{\text{res}}$ being the MLE of ξ_0 under the restricted model, then

$$avar\left[\sqrt{n}\,\left\{\widehat{r}_n^{\text{res}}(x) - r(x)\right\}\right] = C^{-2}\left[\mathrm{E}\left[[\{Y - r(x)\}g_{0,x,\xi_0}(Y)]^2\right] + D\Pi_{\xi_0}V_{\xi_0}^{-1}\Pi_{\xi_0}^\top D^\top\right]. \tag{24}$$

where $\Pi_{\xi_0} = G(G^\top V_{\xi_0}G)^{-1}G^\top V_{\xi_0}$ with

$$G = \begin{pmatrix} \mathbf{I}_{k_1} & 0 & 0 & 0 \\ 0 & B_0^\top \otimes \mathbf{I}_{k_1} & \mathbf{I}_r \otimes \mathbf{A}_0 & 0 \\ 0 & 0 & 0 & \mathbf{I}_{k_2} \end{pmatrix}$$

and $\xi_0 = (\bar{\eta}_{10}, \beta_0, \bar{\eta}_{20})$, with $\beta_0 = \mathbf{A}_0 B_0$, $\mathbf{A}_0 \in \mathbb{R}^{k_1\times d}$ and $B_0 \in \mathbb{R}^{d\times r}$.

Remark 3. Given that Π_{ξ_0} is a projection, we have $\Pi_{\xi_0}V_{\xi_0}^{-1}\Pi_{\xi_0}^\top \leq V_{\xi_0}^{-1}$ and therefore the asymptotic variance of $\sqrt{n}\,\{\widehat{r}_n^{\text{res}}(x) - r(x)\}$, presented in (24) is smaller than avar$[\sqrt{n}\,\{\widehat{r}_n(x) - r(x)\}]$, exhibited in (22).

Remark 4. For the case of $\mathbf{X}|Y = y \sim \mathcal{N}(\mu + \beta\nu_y, \Delta)$ with $\mathrm{E}(\nu_y) = 0$, V_{ξ_0} presented in (23) is given by

Table 1
Average variance and squared bias of the four estimators for different values of the sample size n.

	$n = 200$		$n = 500$		$n = 750$		$n = 1000$		$n = 2000$	
	Sqbias	Var	Sqbias	Var	Sqbias	Var	Sqbias	Var	Sqbias	Var
Inverse-Red	0.013	0.186	0.003	0.047	0.002	0.041	0.001	0.026	0.001	0.015
Inverse	0.023	1.287	0.006	0.646	0.005	0.444	0.002	0.356	0.002	0.182
Multinomial	0.042	0.958	0.042	0.352	0.028	0.194	0.011	0.145	0.033	0.081
RF	0.544	0.150	0.439	0.104	0.401	0.085	0.327	0.077	0.357	0.057

$$V_{\xi_0} = \begin{pmatrix} \Delta & 0 & -(\mu^\top \Delta \otimes \Delta)D_p \\ 0 & \mathrm{E}(\nu_y \nu_y^\top) \otimes \Delta & -\mathrm{E}(\nu_y \nu_y^\top)(\beta^\top \Delta \otimes \Delta)D_p, \\ -D_p^\top(\Delta\mu \otimes \Delta) & -D_p^\top(\Delta\beta\mathrm{E}(\nu_y \nu_y^\top) \otimes \Delta) & W \end{pmatrix}$$

where $W = D_p^\top[\Delta \mathrm{E}\{(\mu + \beta\nu_y)(\mu + \beta\nu_y)^\top\}\Delta \otimes \Delta]D_p + D_p^\top(\Delta \otimes \Delta)D_p/2$.

5. Simulated data

We report the results of a Monte Carlo simulation study comparing the empirical bias and variance of several estimators when the inverse regression of $\mathbf{X}$ on Y follows a reduced rank generalized linear model. In detail, Y is a discrete random variable taking values in $\{1, \ldots, r\}$ with $r = 6$ and probability $1/r$ for each point. Then $\mathbf{X} \mid Y = y \sim \mathcal{N}(\mu_y, \Delta)$ with $\Delta = I_p$, $p = 30$ and $\mu_y = \beta f_y$ for $\beta = \mathbf{A}B$, with $\beta \in \mathbb{R}^{p\times(r-1)}$, $\mathbf{A} = \mathbf{1}_p/\sqrt{p} \in \mathbb{R}^{p\times 1}$ and $B = 5\mathbf{1}_{r-1}{}^\top \in \mathbb{R}^{1\times(r-1)}$; $f_y \in \mathbb{R}^{r-1}$ and its kth coordinate is given by $f_{yk} = \mathbf{1}(y = k) - 1/r$ for $k \in \{1, \ldots, r-1\}$. In practice, the center term $1/r$ is replaced by its estimator n_k/n, where n_k is the cardinal of the set $\{y = k\}$ and n is the sample size.

We consider the following values for the sample size n: 200, 500, 750, 1000, 2000. For each value of n we generate 200 independent training sets of size n. Additionally, we generate one independent testing set size 1000: $x_{test,j}$ for $j \in \{1, \ldots, 1000\}$.

For each prediction procedure considered we compute an estimate of the regression function $\mathrm{E}(Y|\mathbf{X} = x)$, say $\hat{\mathrm{E}}_i(Y|\mathbf{X} = x)$ based on the ith replication of the training set. The variance and squared bias of the prediction procedure at a point $x_{test,j}$ for $j \in \{1, \ldots, 1000\}$ are estimated as

$$\mathrm{var}_j = \frac{1}{200}\sum_{i=1}^{200}\left\{\hat{\mathrm{E}}_i(Y|\mathbf{X} = x_{test,j}) - \frac{1}{200}\sum_{i=1}^{200}\hat{\mathrm{E}}_i(Y|\mathbf{X} = x_{test,j})\right\}^2$$

and

$$\mathrm{sqbias}_j = \left\{\frac{1}{200}\sum_{i=1}^{200}\hat{\mathrm{E}}_i(Y|\mathbf{X} = x_{test,j}) - \mathrm{E}(Y|\mathbf{X} = x_{test,j})\right\}^2$$

respectively. Here, $\mathrm{E}(Y|\mathbf{X} = x_{test,j})$ is computed according to (1). We then compute the average squared bias and variance over the testing set:

$$\mathrm{ave(var)} = \frac{1}{1000}\sum_{j=1}^{1000}\mathrm{var}_j, \quad \mathrm{ave(sqbias)} = \frac{1}{1000}\sum_{j=1}^{1000}\mathrm{sqbias}_j.$$

We consider the following competitors

(a) The estimate defined in (17), Inverse-Red in the table below. The reduced parameter is estimated using the `rrr` R package.
(b) The estimate defined in (4), Inverse in the table below.
(c) A forward multinomial regression model, computed using the `nnet` R package.
(d) The Random Forest algorithm from [4], computed using the `randomForest` R package, RF in the table below.

Table 1 shows the results. We see that Inverse-Red has a much higher variance than Inverse and a slightly lower bias. The variance of RF is similar to that of Inverse-Red for $n = 200$ but approximately double that of Inverse-Red for larger values of n. Its bias is orders of magnitude larger than those of the methods based on the inverse regression model. This is to be expected, since the data generating process follows the inverse generalized linear model exactly, whereas Random Forest is a nonparametric technique. In contrast, the variance of Multinomial is larger than that of Inverse-Red in all cases and its bias is always larger than those of the methods based on inverse regression, but smaller than that of the RF method.

6. Example

We consider the Australian Athletes example from [6]. Lean body mass (LBM), Y, is regressed on $\mathbf{X} = (W, \mathbf{Z}) \in \mathbb{R}^p$ with $p = 6$, where W is an indicator for gender and $\mathbf{Z}$ is a 5-dimensional vector with the logarithms of height, weight, red cell count, white cell count and hemoglobin, for a sample of $n = 202$ individuals training at the Australian Institute of Sport. To apply the dimension reduction methodology developed in [6] and also considered in this work, it is required to model the inverse regression $\mathbf{X}|Y$.

The plots of the $\mathbf{Z}$ component versus Y at each level of W suggest that no interaction between $\mathbf{X}$ and W is required to model $\mathrm{E}\{\mathbf{Z}|(W, Y)\}$ as a linear function of Y, except for a different intercept for each sex, i.e., $\mathrm{E}\{\mathbf{Z}|(W, Y)\} = \mu + bW + c\{Y - \mathrm{E}(Y)\}$. Bura et al. [6] model $\mathbf{Z}|(W, Y)$ as a multivariate normal with mean $\mathrm{E}\{\mathbf{Z}|(W, Y)\}$ and constant variance. They also assume a logistic model for $W|Y$ as a function of Y. Combining the postulated models for $\mathbf{Z}|(W, Y)$ and for $W|Y$, the distribution of $\mathbf{X}|Y$ follows an exponential family as in the example of Section 2.1 where the shape of the parameters for $\mathbf{Z}|(W, Y)$ and $W|Y$, which are related with $\eta = \eta_y = (\eta_{1,y}, \eta_{2,y}^\top, \eta_3^\top, \eta_4^\top, \eta_5^\top)^\top$ through Eq. (8), should be modeled in such a way that $(\eta_{1,y}, \eta_{2,y}^\top)^\top$ depends linearly on $\nu_y = y - \mathrm{E}(Y)$, i.e.,

$$(\eta_{1,y}, \eta_{2,y}^\top)^\top = \overline{\eta}_1 + \beta\nu_y,$$

where $\overline{\eta}_1 \in \mathbb{R}^p$ and $\beta \in \mathbb{R}^{p\times 1}$. As a consequence, Bura et al. [6] proved that a sufficient reduction for the regression of Y on $\mathbf{X}$ is $R(x) = \beta^\top(w, z^\top)^\top$. After computing the estimator of the reduction, these authors plot the response, LBM, vs. the estimated sufficient reduction. The plot indicates that a quadratic function in $R(\mathbf{X})$ can be fitted to predict LBM based on the estimated reduction $\hat{R}(\mathbf{X})$. In Section 6.2 of [6], the authors compared the predictive performance of several methods including Kernel Dimension Reduction [20], Partial SIR [10], the procedure discussed in this paper, and a quadratic regression model on $\hat{R}(\mathbf{X})$. They computed the leave-one-out prediction mean square error and concluded that the best results were obtained using the quadratic regression model. However, no theory is available to compute confidence intervals for the regression function based on this approach. Instead, Theorem 4 can be used to construct asymptotic confidence intervals for $r(x)$, both under the unrestricted and the restricted models. Actually, Eqs. (22) and (24) establish that $\widehat{r}_n(x)$ and $\widehat{r}_n^{\mathrm{res}}(x)$ are asymptotically normal and an explicit formula for their asymptotic variance is provided. Note that, in the present example, no rank restriction is assumed because β itself has rank $d = 1$ and so, we only use $\widehat{r}_n(x)$.

A plug-in procedure can be used to estimate the asymptotic variance presented on the left-hand side of (22) with $\widehat{\mathrm{avar}}$, and therefore, appealing to Slutzky's Theorem, asymptotic confidence intervals for $r(x)$ can be constructed as

$$\mathrm{IC}_{\mathrm{Slu}}\{r(x)\} = \widehat{r}_n(x) \pm z_{\alpha/2}\sqrt{\mathrm{var}_{\mathrm{Slu}}\{\widehat{r}_n(x)\}},$$

where $\mathrm{var}_{\mathrm{Slu}}\{\widehat{r}_n(x)\} = \widehat{\mathrm{avar}}_{\mathrm{Slu}}/n$.

Also, a bootstrap variance estimation can be computed, sampling with replacement from the empirical distribution, giving rise to the so-called normal bootstrap intervals, presented in Section 8.3 of [33], defined as

$$\mathrm{IC}_{\mathrm{Boot}}(r(x)) = \widehat{r}_n(x) \pm z_{\alpha/2}\sqrt{\mathrm{var}_{\mathrm{Boot}}\{\widehat{r}_n(x)\}}.$$

To illustrate possible applications of the results presented in this work, we compute confidence intervals according to each of the described methods for $r^{-i}(\mathbf{X}_i)$, for each $i \in \{1, \ldots, 202\}$, where r^{-i} stands for the prediction estimated using all the observations in the sample, except for observation number i. Fig. 1 shows the results of the 95% confidence intervals computed with both methods (asymptotic in red and bootstrap in blue). For the central values of the estimated reduction both methods perform similarly. However, as expected, the variability increases when the estimated reduction is away from the center. Intuitively, this is due to there being more information near the mean of estimated reduction.

In many situations, it may also be desirable to construct prediction intervals for the response at a fixed value x of the predictors. Namely, given $\alpha \in (0, 1)$, we look for a random interval $(A_{n,\alpha}, B_{n,\alpha})$ constructed with $(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)$ such that

$$\Pr\left(A_{n,\alpha} \le Y_{new} \le B_{n,\alpha}|\mathbf{X}_{new} = x\right) \to 1 - \alpha \quad \text{a.s.}$$

These intervals can be constructed easily provided $F_{Y|\mathbf{X}=x}$, the conditional distribution of Y given $\mathbf{X} = x$, can be consistently estimated. In such a case, the $\alpha/2$ and $1 - \alpha/2$ quantiles of the estimated conditional distribution satisfy the definition of prediction intervals.

To estimate $F_{Y|\mathbf{X}=x}(y) = \Pr(Y \le y|\mathbf{X} = x)$ it is useful to represent it as $F_{Y|\mathbf{X}=x}(y) = \mathrm{E}\{\mathbf{1}(Y \le y)|\mathbf{X} = x\}$ and, appealing again to Bayes' Theorem as in Eq. (1), we get that $F_{Y|\mathbf{X}=x}(y) = \mathrm{E}\{\mathbf{1}(Y \le y)f(x|Y)\}/\mathrm{E}\{f(x|Y)\}$. Mimicking the construction of the estimators of $r(x)$ developed in this work, we arrive at the following estimators of $F_{Y|\mathbf{X}=x}(y)$

$$\widehat{F}_{Y|\mathbf{X}=x}(y) = \sum_{i=1}^n \mathbf{1}(Y_i \le y)f(x|Y_i, \widehat{\xi}_n)\Big/\sum_{i=1}^n f(x|Y_i, \widehat{\xi}_n), \quad \widehat{F}^{\mathrm{res}}_{Y|\mathbf{X}=x}(y) = \sum_{i=1}^n \mathbf{1}(Y_i \le y)f(x|Y_i, \widehat{\xi}_n^{\mathrm{res}})\Big/\sum_{i=1}^n f(x|Y_i, \widehat{\xi}_n^{\mathrm{res}}).$$

That is, both $\widehat{F}_{Y|\mathbf{X}=x}$ and $\widehat{F}^{\mathrm{res}}_{Y|\mathbf{X}=x}$ are concentrated in $Y_1, \ldots, Y_n$, and assign weights

$$w_i(x) = f(x|Y_i, \widehat{\xi}_n)\Big/\sum_{j=1}^n f(x|Y_j, \widehat{\xi}_n) \quad \text{and} \quad w_i(x) = f(x|Y_i, \widehat{\xi}_n^{\mathrm{res}})\Big/\sum_{j=1}^n f(x|Y_j, \widehat{\xi}_n^{\mathrm{res}}), \tag{25}$$

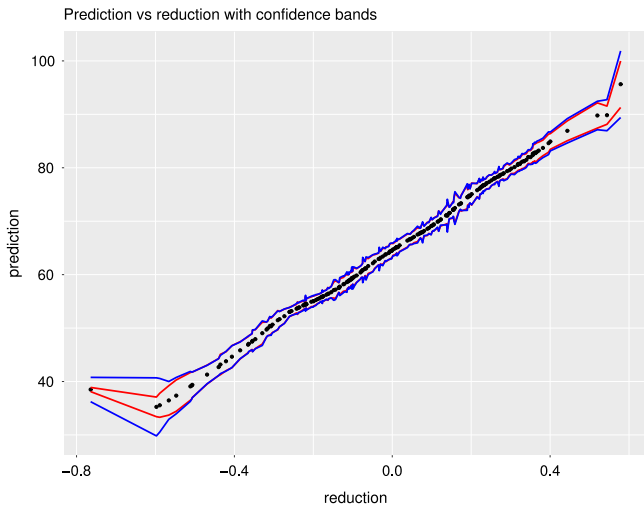

Fig. 1. LBM prediction versus estimated sufficient reduction. Confidence intervals for $r(x)$: asymptotic in red and bootstrap in blue . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

respectively. As a matter of fact, the regression estimators presented in Eqs. (12) and (17) coincide with the mean operator evaluated at $\widehat{F}_{Y|\mathbf{X}=x}$ and $\widehat{F}^{\text{res}}_{Y|\mathbf{X}=x}$, respectively.

To evaluate empirically the performance of this procedure, we use again the Australian Athletes example. For each observation $i \in \{1, \ldots, 202\}$, we computed a 95% prediction interval considering $\mathbf{X}_{new} = \mathbf{X}_i$ and using the 201 observations $(\mathbf{X}_j, Y_j)$ with $j \neq i$. In 95.05% of the cases, the constructed intervals contained the observed response Y_i, supporting the theoretical considerations discussed above.

7. Discussion and future work

We have proved the consistency and asymptotic normality of the prediction based on a sufficient dimension reduction when a generalized linear model is postulated for the inverse regression. The results are illustrated in a real data example, where confidence intervals for the regression function are computed. We also present prediction intervals, estimating $F_{Y|\mathbf{X}=x}$. Interestingly, once an estimate of $F_{Y|\mathbf{X}=x}$ is available, we can then estimate percentiles, but also estimate any parameter associated with the conditional distribution defined through a continuous functional. For instance, we can estimate α-trimmed means and more general L-functionals as well as M-location and scale functionals; see [28]. These parameters are typically of interest to the robust statistics community, in the sense that, for a univariate data set, they are not strongly affected by small deviations from the model assumptions. However, in the present setting, to keep such a robust property, a robust fit should also be considered in the estimation of ξ_0. The development of these procedures is part of a work in progress.

Single index models are an alternative way of achieving dimension reduction. For instance, the single index model for the conditional mean establishes that $\mathrm{E}(Y \mid X = x) = G(\beta_0^\top x)$, where $G : R \to R$ is an unknown function. Other authors postulate that $Y = G(\beta_0^\top x) + \varepsilon$, with ε independent of X and $\mathrm{E}(\varepsilon) = 0$. In such a case, the conditional distribution of $Y \mid X$ depends on X only through $\beta_0^\top X$; namely, $Y \mid X \sim Y \mid \beta_0^\top X$ and, therefore, $\beta_0^\top X$ is a sufficient reduction for the regression of Y on X. Importantly, unlike the approach discussed in this paper, single index models require a model for the forward regression function. It is known that under regularity assumptions (see Chapter 2 in [21]), β_0 can be estimated (except for a multiplicative constant) at the same rate as that available in the parametric model in which G is known ($n^{-1/2}$), while regression kernels methods (such as Nadaraya–Watson) can be used to estimate G at the same rate in the model where the one-dimensional quantity $\beta_0^\top X$ is observable. Such results suggest that the response for a newly observed x can be predicted with $\widehat{G}(\widehat{\beta}^\top x)$. Studying the theoretical properties of these procedure is an interesting problem, that could be considered an extension of the results presented in this paper. We are not aware of any results in this direction. The previous discussion is also valid for the multiple index model, where the dimension reduction is given by $\beta_0 X$ and now β_0 is a matrix with less rows than columns.

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Technical details

The proofs of the results stated in these paper are available online in the Online Supplement http://doi.org/10.1016/j.jmva.2018.12.003.

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